

Problem-Solution Fit

Date	15 March 2026
Team ID	-
Project Name	Flood Prediction Using Machine Learning
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

- 1. Customer Segment(s):** Disaster Management Authorities, Government Agencies, Local Administrations, Residents in flood-prone areas.
- 2. Problems / Pains:** Late flood prediction, inaccurate forecasting, property damage, loss of lives, insufficient early warning.
- 3. Triggers to Act:** Heavy rainfall, rising river levels, extreme weather forecasts.
- 4. Emotions:** Before: Fear, uncertainty. After: Confidence, preparedness, safety.
- 5. Available Solutions:** Traditional forecasting and manual monitoring; limited accuracy and slower response.
- 6. Customer Limitations:** Limited budget, limited access to sensors/devices, incomplete historical data.
- 7. Behavior:** Authorities monitor weather data daily and issue alerts during monsoon.
- 8. Channels of Behavior:** Online: Web dashboard, SMS/Email alerts. Offline: Emergency response teams, public announcements.
- 9. Problem Root Cause:** Complex weather patterns and inability of traditional methods to analyze multiple factors accurately.
- 10. Your Solution:** An ML model predicts flood risk using rainfall, river level and historical weather data, providing early warnings and supporting disaster management decisions.